Introductory activity

Activity 1 · The solutions of $z^n = 1$ 40 min.

1. Factorise $z^3 - 1$ by $z - 1$, then solve $z^3 = 1$ in $\mathbb{C}$. Let j be the solution with positive imaginary part.
2. Plot the three solutions in the complex plane. Compute the distances between them. Which figure do you obtain?
3. Write j in exponential form. Check that $j^2 = \overline{j}$ and that $1 + j + j^2 = 0$. Interpret the last equality with the centroid.
4. Solve $z^4 = 1$ and plot the solutions. Same question for $z^6 = 1$, using $z^6 - 1 = \left(z^3 - 1\right)\left(z^3 + 1\right)$.
5. State a conjecture about the solutions of $z^n = 1$: their number, their modulus, their arguments, the figure they form.

Observing is not proving: the properties conjectured here are established in the course.